

Main tables and figures

Locally robust commensurate Bayesian additive regression trees for historical information borrowing

Tables 1–7 and Figures 1–2 accompany the main manuscript. Full configurations, Monte Carlo standard errors and implementation details appear in the separate appendix.

Table 1: Simulation study designs.

Design	Trial N	Historical n_3	$P(Z = 1)$	Replicates	Settings
Gaussian, two-arm	300	300	$2/3$	100	Sc1–Sc5; 12 settings
Survival, two-arm	300	300	$3/4$	100	Sc1–Sc5; 12 settings
Gaussian, single-arm	200	300	1	100	Sc1–Sc2
Survival, single-arm	30	300	1	100	Sc1–Sc2
Survival, single-arm	200	300	1	100	Sc1–Sc2

Ten predictors are generated in each design. $Z = 1$ denotes trial treatment; two-arm allocation is Bernoulli, so realized arm sizes vary. Historical controls are regenerated for each replicate. Sc1: common covariate distributions; Sc2: selection on observed prognostic variables; Sc3: latent-variable selection with proxy correlations $-0.5, 0, 0.5$; Sc4: global historical shift of 1 or 2; Sc5: regional shifts outside $X_5 > 2$ (0.5, 1, 2) or $X_7 \leq 2$ (1, 2). Gaussian treatment effects are 1; survival effects are $\log(1.4)$ on the log-time scale. The two single-arm survival sample sizes belong to one study.

Table 2: Gaussian mean-difference performance in representative two-arm scenarios.

Method	Bias	RMSE	Coverage	Power
<i>Sc1</i>				
lrcBART	0.003	0.174	0.950	0.890
BART-NP	-0.010	0.209	0.970	0.730
BART-CP	0.009	0.153	0.950	0.980
BART-PP	-0.001	0.197	0.950	0.800
LM-NP	-0.019	0.348	0.930	0.390
LM-CP	0.015	0.262	0.940	0.710
LM-PP	-0.005	0.349	0.950	0.390
hier. LM	-0.002	0.329	0.940	0.450
PSCL	0.019	0.290	0.920	0.480
MAP	0.023	0.302	0.940	0.530
<i>Sc2</i>				
lrcBART	0.003	0.186	0.940	0.860
BART-NP	-0.032	0.211	0.940	0.750
BART-CP	0.048	0.188	0.930	0.950
BART-PP	-0.004	0.200	0.970	0.740
LM-NP	-0.062	0.358	0.900	0.440
LM-CP	0.343	0.463	0.730	0.920
LM-PP	0.007	0.355	0.950	0.450
hier. LM	0.052	0.363	0.930	0.510
PSCL	0.007	0.317	0.910	0.390
MAP	0.186	0.458	0.870	0.550
<i>Sc3, $\rho = 0$</i>				
lrcBART	0.009	0.315	0.930	0.560
BART-NP	-0.046	0.328	0.950	0.520
BART-CP	0.957	0.985	0.010	1.000
BART-PP	0.004	0.309	0.940	0.550
LM-NP	-0.057	0.367	0.930	0.340
LM-CP	0.989	1.027	0.050	1.000
LM-PP	0.047	0.350	0.940	0.500
hier. LM	0.085	0.373	0.930	0.510
PSCL	0.667	0.721	0.290	0.980
MAP	0.202	0.434	0.910	0.610

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Table 2 (continued)

Method	Bias	RMSE	Coverage	Power
<i>Sc4</i> , $\delta = 1$				
lrcBART	-0.007	0.201	0.940	0.810
BART-NP	-0.010	0.209	0.970	0.730
BART-CP	-0.689	0.706	0.000	0.000
BART-PP	-0.012	0.197	0.970	0.770
LM-NP	-0.019	0.348	0.930	0.390
LM-CP	-0.726	0.772	0.180	0.010
LM-PP	-0.054	0.352	0.960	0.350
hier. LM	-0.104	0.358	0.930	0.290
PSCL	-0.483	0.563	0.530	0.060
MAP	-0.230	0.467	0.860	0.170
<i>Sc5</i> , X_5 , $\delta = 1$				
lrcBART	-0.050	0.214	0.930	0.690
BART-NP	-0.010	0.209	0.970	0.730
BART-CP	-0.334	0.366	0.370	0.350
BART-PP	-0.014	0.195	0.970	0.760
LM-NP	-0.019	0.348	0.930	0.390
LM-CP	-0.353	0.439	0.660	0.190
LM-PP	-0.030	0.348	0.970	0.370
hier. LM	-0.063	0.338	0.920	0.360
PSCL	-0.230	0.370	0.850	0.210
MAP	-0.160	0.378	0.920	0.270

Each row summarizes 100 alternative-hypothesis replicates. Coverage is for a 95% interval. Empirical power is the frequency of posterior probability of benefit exceeding 0.95 (PSCL: lower 95% confidence limit above the threshold). Gaussian decisions use a threshold of 0.5 and survival decisions use a ratio threshold of 1. RMSE is the square root of the mean squared error. NP denotes no historical pooling (trial controls only); CP denotes complete pooling; PP denotes source-adjusted pooling using a source indicator. PP is distinct from the hierarchical comparator. lrcBART requests ESS 100; the MAP comparator in Table 2 also requests 100. Full configurations and Monte Carlo standard errors appear in the appendix. The hierarchical comparator uses prior setting $a = 0.25$.

Table 3: Median-survival-ratio performance in representative two-arm scenarios.

Method	Bias	RMSE	Coverage	Power
<i>Sc1</i>				
lrcBART	0.014	0.290	0.970	0.420
BART-NP	0.048	0.392	0.980	0.330
BART-CP	0.010	0.218	0.950	0.660
BART-PP	-0.034	0.316	0.980	0.300
AFT-NP	-0.105	0.455	0.970	0.070
AFT-CP	0.027	0.307	1.000	0.330
AFT-PP	0.022	0.415	0.990	0.100
hier. AFT	0.050	0.443	0.990	0.140
<i>Sc2</i>				
lrcBART	0.058	0.341	0.960	0.480
BART-NP	0.117	0.440	0.960	0.320
BART-CP	0.130	0.310	0.970	0.630
BART-PP	-0.019	0.349	0.970	0.280
AFT-NP	0.018	0.529	0.960	0.160
AFT-CP	1.202	1.435	0.570	0.920
AFT-PP	0.403	0.817	0.920	0.300
hier. AFT	0.417	0.815	0.930	0.390
<i>Sc3, $\rho = 0$</i>				
lrcBART	0.076	0.472	0.970	0.290
BART-NP	0.014	0.502	0.960	0.180
BART-CP	2.642	2.802	0.000	1.000
BART-PP	0.075	0.446	0.960	0.300
AFT-NP	-0.090	0.531	0.970	0.110
AFT-CP	3.187	3.429	0.010	1.000
AFT-PP	0.477	0.839	0.930	0.470
hier. AFT	0.415	0.836	0.920	0.350

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Table 3 (continued)

Method	Bias	RMSE	Coverage	Power
<i>Sc4</i> , $\delta = 1$				
lrcBART	-0.007	0.332	0.940	0.370
BART-NP	0.048	0.392	0.980	0.330
BART-CP	-0.729	0.736	0.000	0.000
BART-PP	-0.080	0.311	0.960	0.230
AFT-NP	-0.105	0.455	0.970	0.070
AFT-CP	-0.768	0.780	0.080	0.000
AFT-PP	-0.212	0.405	0.960	0.030
hier. AFT	-0.159	0.459	0.970	0.060
<i>Sc5</i> , X_5 , $\delta = 1$				
lrcBART	-0.054	0.331	0.930	0.330
BART-NP	0.048	0.392	0.980	0.330
BART-CP	-0.391	0.422	0.420	0.070
BART-PP	-0.067	0.316	0.960	0.240
AFT-NP	-0.105	0.455	0.970	0.070
AFT-CP	-0.441	0.484	0.700	0.000
AFT-PP	-0.116	0.385	1.000	0.060
hier. AFT	-0.074	0.433	0.980	0.080

Each row summarizes 100 alternative-hypothesis replicates. Coverage is for a 95% interval. Empirical power is the frequency of posterior probability of benefit exceeding 0.95 (PSCL: lower 95% confidence limit above the threshold). Gaussian decisions use a threshold of 0.5 and survival decisions use a ratio threshold of 1. RMSE is the square root of the mean squared error. NP denotes no historical pooling (trial controls only); CP denotes complete pooling; PP denotes source-adjusted pooling using a source indicator. PP is distinct from the hierarchical comparator. lrcBART requests ESS 100; the MAP comparator in Table 2 also requests 100. Full configurations and Monte Carlo standard errors appear in the appendix. The hierarchical comparator uses prior setting $a = 0.25$.

Table 4: Single-arm performance for Gaussian mean differences and median-survival ratios.

Method	Bias	RMSE	Coverage	Power
<i>Gaussian, Sc1</i>				
lrcBART	-0.015	0.156	0.950	0.890
BART-CP	-0.014	0.153	0.940	0.940
LM-CP	-0.029	0.270	0.960	0.600
hier. LM	-0.025	0.272	1.000	0.090
hier. LM	-0.026	0.320	1.000	0.000
<i>Gaussian, Sc2</i>				
lrcBART	0.062	0.177	0.980	0.910
BART-CP	0.056	0.176	0.930	0.950
LM-CP	0.290	0.395	0.850	0.910
hier. LM	0.357	0.444	1.000	0.500
hier. LM	0.293	0.433	1.000	0.000
<i>Survival, Sc1</i>				
lrcBART	-0.020	0.178	0.990	0.590
BART-CP	-0.011	0.180	0.970	0.730
AFT-CP	0.055	0.373	0.970	0.370
hier. AFT	0.097	0.388	1.000	0.020
hier. AFT	0.065	0.377	1.000	0.000
<i>Survival, Sc2</i>				
lrcBART	0.070	0.216	0.980	0.730
BART-CP	0.069	0.216	0.990	0.770
AFT-CP	0.676	0.859	0.820	0.810
hier. AFT	0.872	1.042	1.000	0.370
hier. AFT	0.722	0.908	1.000	0.000

Each row summarizes 100 alternative-hypothesis replicates. Coverage is for a 95% interval; empirical power is the frequency of posterior probability of benefit exceeding 0.95, using a threshold of 0.5 for Gaussian mean differences and 1 for survival ratios. All single-arm lrcBART primary rows use fixed $w = 1$, $H_f = 50$, $H_g = 5$ and requested ESS 100. The hierarchical variance prior is $IG(3/2, 3a/2)$. In single-arm studies, CP uses all historical controls with full likelihood weight and no source discrepancy; there are no observed trial controls to pool. Full configurations and Monte Carlo standard errors appear in the appendix. Within each outcome and scenario, the first and second hier. LM or hier. AFT rows use prior settings $a = 0.05$ and $a = 0.5$, respectively.

Table 5: Characteristics of the harmonized multiple-myeloma analysis cohorts.

Characteristic	EloKRd	UCMM
Participants	30	253
Age (years), median (IQR)	62.8 (57.4, 70.2)	61.3 (54.4, 67.6)
Male	21 (70.0)	130 (51.4)
Race		
White	23 (76.7)	164 (64.8)
Black	4 (13.3)	76 (30.0)
Other	3 (10.0)	13 (5.1)
Hispanic or Latino	1 (3.3)	13 (5.1)
High-risk cytogenetics	15 (50.0)	58 (22.9)
ASCT	9 (30.0)	203 (80.2)
Progression or death	13 (43.3)	159 (62.8)
All-cause death	10 (33.3)	60 (23.7)

Entries are n (%) except where indicated. IQR, interquartile range; ASCT, autologous stem-cell transplantation. ASCT is a postinduction characteristic and is not presented as a pretreatment baseline variable. All analysis covariates were complete in both cohorts.

Table 6: Five-year restricted mean survival time (RMST) in the application.

Method	RMST ratio (95% interval)	RMST difference (95% interval)	EloKRd treatment RMST	EloKRd control RMST	Historical model RMST	Observed UCMM RMST
<i>PFS</i>						
KM	1.015 (0.845, 1.220)	0.055 (-0.625, 0.735)	3.705	—	—	3.649
lrcBART	0.965 (0.783, 1.186)	-0.129 (-0.853, 0.614)	3.524	3.656	3.656	—
BART-CP	0.975 (0.794, 1.154)	-0.090 (-0.772, 0.522)	3.527	3.617	3.617	—
AFT-CP	0.869 (0.695, 1.061)	-0.460 (-1.109, 0.199)	3.069	3.527	3.527	—
hier. AFT	1.034 (0.759, 1.379)	0.109 (-0.965, 0.984)	3.355	3.225	3.591	—
hier. AFT	1.204 (0.733, 1.936)	0.569 (-1.192, 1.640)	3.355	2.744	3.599	—
<i>OS</i>						
KM	0.902 (0.786, 1.034)	-0.450 (-1.019, 0.118)	4.122	—	—	4.572
lrcBART	0.903 (0.774, 1.033)	-0.427 (-1.019, 0.135)	3.990	4.419	4.421	—
BART-CP	0.907 (0.769, 1.018)	-0.411 (-1.045, 0.075)	3.990	4.406	4.406	—
AFT-CP	0.803 (0.649, 0.937)	-0.850 (-1.519, -0.262)	3.449	4.306	4.306	—
hier. AFT	0.963 (0.756, 1.391)	-0.148 (-1.081, 1.122)	3.807	3.949	4.412	—
hier. AFT	1.188 (0.766, 1.688)	0.594 (-1.096, 1.648)	3.807	3.171	4.417	—

RMST and differences are in years. EloKRd control RMST is the estimated restricted mean survival under control for the EloKRd population. Adjusted methods standardize to EloKRd covariates and compare treatment with this estimated control outcome; KM compares the observed EloKRd and UCMM cohorts. Historical-model RMST evaluates the historical-control model at EloKRd covariates, whereas observed UCMM RMST is the unadjusted cohort summary used only by KM. For lrcBART, the EloKRd control uses $f + g$ and the historical-model column uses f , with the respective residual variances. BART-CP and AFT-CP have identical model-based control columns because they have no discrepancy component. A dash denotes an inapplicable estimate. KM is an unadjusted comparison with confidence intervals; model-based intervals are equal-tailed credible intervals. Arm summaries are posterior medians, except KM estimates. The median of a ratio or difference need not equal the ratio or difference of marginal medians. Primary lrcBART uses $H_f = 10$, $H_g = 5$, $w = 1$, requested ESS 30. Within each endpoint, the first and second hier. AFT rows use prior settings $a = 0.05$ and $a = 0.5$, respectively.

Table 7: Observed cohort counts and historical prior ESS for joint covariate combinations observed in EloKRd (PFS).

Age band	Sex	Race	Hispanic/ Latino	Cytogenetic risk	ASCT	EloKRd n	UCMM n	Historical prior ESS
< 50	Male	White	No	High	No	1	0	9.0
< 50	Male	White	No	High	Yes	1	1	12.1
< 50	Male	Black	No	Standard	No	1	0	10.9
50–59	Female	White	No	Standard	No	1	2	13.3
50–59	Female	White	No	Standard	Yes	1	17	19.7
50–59	Female	White	No	High	No	1	1	9.4
50–59	Female	Black	No	High	No	1	0	8.1
50–59	Male	White	No	Standard	No	1	3	13.6
50–59	Male	White	No	Standard	Yes	3	20	23.4
50–59	Male	White	No	High	Yes	1	8	15.6
50–59	Male	Other	No	Standard	Yes	1	1	8.5
60–69	Female	White	No	Standard	No	1	3	13.4
60–69	Female	White	No	High	No	1	2	10.1
60–69	Male	White	No	Standard	Yes	1	22	21.6
60–69	Male	White	No	High	No	3	2	11.0
60–69	Male	Other	No	High	No	1	0	6.2
60–69	Male	Other	Yes	Standard	No	1	0	5.4
≥ 70	Female	White	No	High	No	1	1	9.4
≥ 70	Female	Black	No	Standard	No	2	7	11.8
≥ 70	Male	White	No	Standard	No	2	5	15.0
≥ 70	Male	White	No	High	No	3	1	10.1
≥ 70	Male	White	No	High	Yes	1	2	12.9

Only the 22 combinations observed in EloKRd are shown, ordered by the covariate columns. Counts are observed patients within each combination. The displayed counts sum to 30 EloKRd and 98 UCMM patients; other UCMM combinations are omitted from this table. Each ESS uses the full 253-patient UCMM fit, exact trial ages, and the same all-spike calibration law and globally selected PFS scale. Units are equivalent uncensored normal-reference controls for mean control log-PFS. Subgroup ESS values are not additive and do not measure outcome compatibility. ASCT is postinduction.

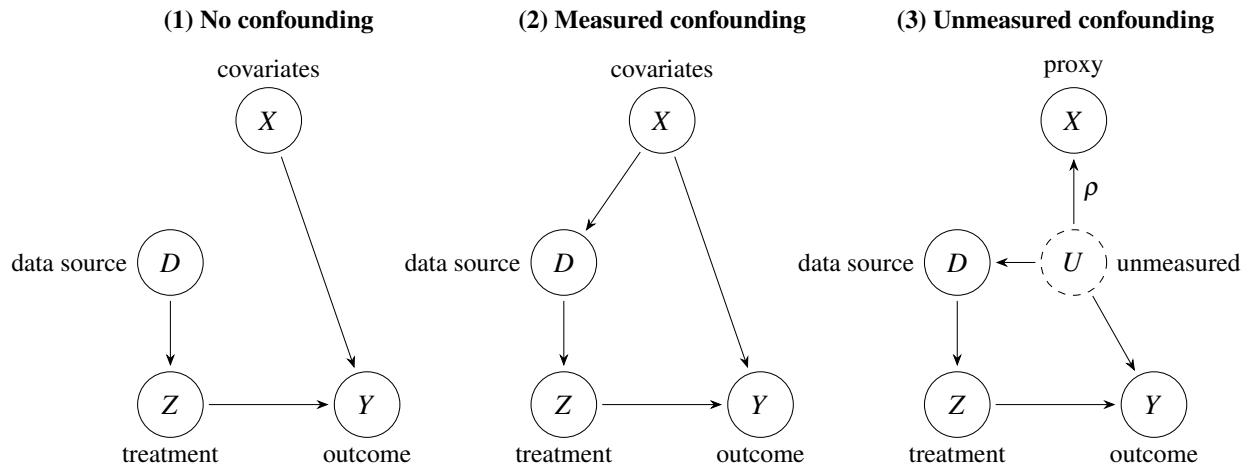


Figure 1: Directed acyclic graphs (DAGs) for the source-selection mechanisms in Sc1–Sc3. In this schematic, D denotes data source (trial or historical), Z treatment assignment, Y an individual outcome, X the measured predictors relevant to the depicted paths and U unmeasured prognostic variables; participant and group indices are suppressed. Sc1: measured covariates predict the outcome but do not influence source selection. Sc2: measured covariates influence both source selection and outcome. Sc3: unmeasured variables influence source selection and outcome, and generate observed proxies through $U \rightarrow X$, with proxy association indexed by ρ . Other measured predictors are omitted from the schematic. The $D \rightarrow Z$ edge reflects that historical observations are controls, whereas treatment is assigned within the trial. These diagrams illustrate the simulation mechanisms, not an established causal structure for the clinical application.

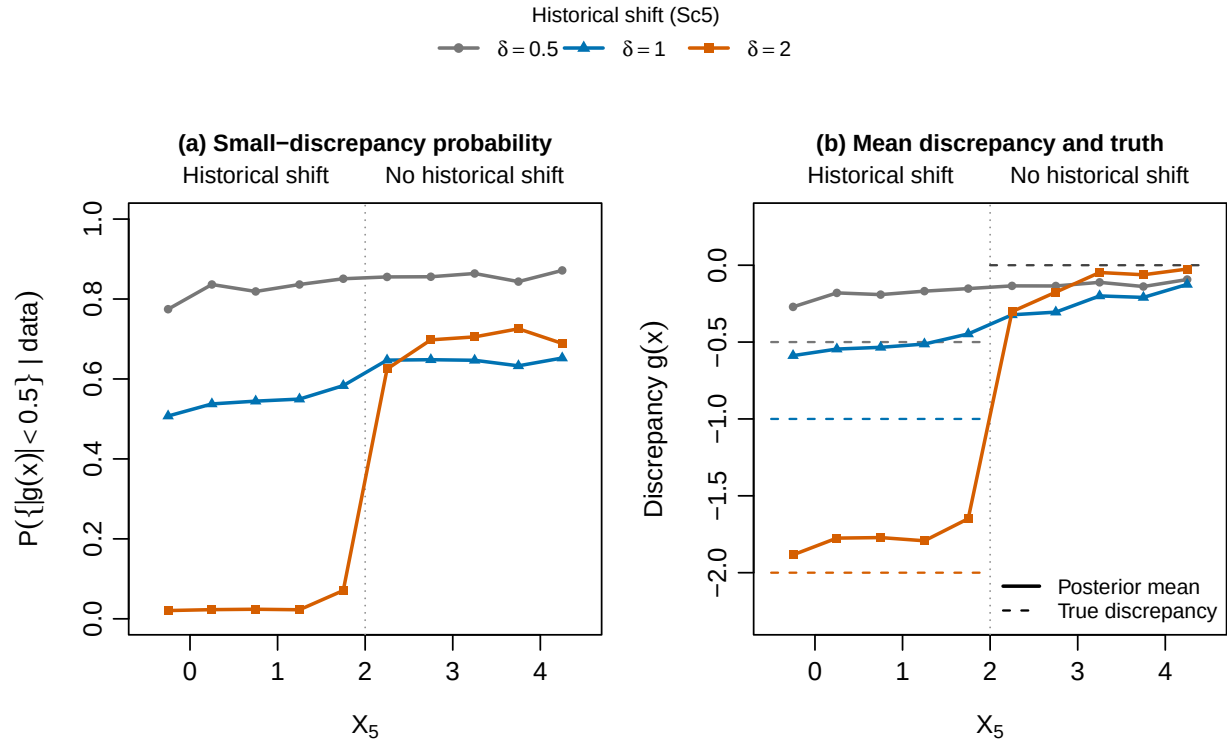


Figure 2: Regional historical shifts for Gaussian outcomes (Sc5). Historical outcomes are shifted upward by $\delta = 0.5, 1, 2$ where $X_5 \leq 2$, and are unshifted otherwise. (a) Posterior probability of a small discrepancy, $P(\{|g(x)| < 0.5\} | \text{data})$. (b) Posterior mean discrepancy; dashed lines show the true trial-minus-historical discrepancy, $-\delta$ in the shifted region and zero elsewhere. Solid curves summarize pooled trial-control profiles within X_5 bins of width 0.5 across 40 replicates; they are not uncertainty intervals. The dotted line marks the shift boundary. Primary lrcBART uses $H_f = 50$, $H_g = 5$, $w \sim \text{Beta}(1, 1)$ and requested prior ESS 100.